Practical1

```
print("Hello World")
```

How to run it:

Open Notepad (or any text editor).

Copy the above code and save the file as hello.py (make sure the extension is .py).

Open Command Prompt (Windows) or

Practical 3

```
<!DOCTYPE html>
<html>
<body>

<script>
var student = {
   "name": "Samita",
   "rollNo": 101,
   "course": "Computer Engineering"
};

document.write("Name: " + student.name + "<br>");
document.write("Roll No: " + student.rollNo + "<br>");
document.write("Course: " + student.course);
</script>

</body>
```

```
</html>
```

1️⃣ **<!DOCTYPE html>**

- This line tells the **browser** that this document is written in **HTML5**.
- It helps the browser to display the webpage correctly.
- It is **not a tag**, it is an instruction.

✅ Always written at the **top** of the HTML file.

2️⃣ **<html>**

- This is the **root (main) tag** of the HTML document.
- All HTML code is written **inside this tag**.

```
<html>
  <!-- all code goes here -->
</html>
```

3️⃣ **<body>**

- The <body> tag contains **everything that is shown on the web page**.
- Text, images, output, etc. appear inside this tag.

```
<body>
  <!-- visible content -->
</body>
```

4️⃣ **<script>**

- The <script> tag is used to write **JavaScript code**.
- JavaScript is used to make the webpage **dynamic** (interactive).

```
<script>
```

```
  // JavaScript code
</script>
```

5️⃣ var student = { ... };

```
var student = {
  "name": "Samita",
  "rollNo": 101,
  "course": "Computer Engineering"
};
```

Explanation:

- var → used to **declare a variable**
- student → name of the variable
- { } → this is an **object**
- Object stores data in **key : value** pairs

Inside the object:

Key	Value
name	Samita
rollNo	101
course	Computer Engineering

👉 This looks like **JSON**, but inside JavaScript it is called a **JavaScript Object**.

6️⃣ document.write()

What is document?

- document represents the **web page**.

What is write()?

- write() is used to **display output on the web page**.

🔹 **First output line**

```
document.write("Name: " + student.name + "<br>");
```

- "Name: " → text to display
- + → joins text with data
- student.name → accesses the **name** from the object
-
 → line break (new line)

✅ Output:

Name: Samita

🔹 **Second output line**

```
document.write("Roll No: " + student.rollNo + "<br>");
```

- student.rollNo → accesses roll number
-
 → moves to next line

✅ Output:

Roll No: 101

🔹 **Third output line**

```
document.write("Course: " + student.course);
```

- student.course → accesses course name
- No
 → output stays on same line

✅ Output:

Course: Computer Engineering

7️⃣ **</script>**

- Ends the JavaScript code block.

8️⃣ **</body>**

- Ends the visible content of the webpage.

9️⃣ **</html>**

- Ends the HTML document.

🖥️ **Final Output on Browser**

Name: Samita

Roll No: 101

Course: Computer Engineering

student → object name

"name", "rollNo", "course" → properties (keys)

"Samita", 101, "Computer Engineering" → values

Data is stored in key : value format

➡ This is a JSON object

6️⃣ Accessing JSON Properties

Copy code

Javascript

student.name

student.rollNo

student.course

Dot (.) operator is used

It gets the value from the object

7️⃣ Displaying Output

Copy code

Javascript

```
document.write("Name: " + student.name + "<br>");
```

document.write() → prints output on webpage

"+" → joins text and value

 → line break

8️⃣ </script>

Ends JavaScript code